Indices and sets	
$\boldsymbol{I}$	Set of assembly tasks, indexed by i
$\boldsymbol{J}$	Set of equipment types, indexed by j
$\boldsymbol{K}$	Set of workstations, indexed by k
$\boldsymbol{P}$	Set of precedence relations i,h, where task i must precede task h
Parameters	
$\boldsymbol{t_{ij}}$	Processing time of task $\boldsymbol{i}$ on equipment type $\boldsymbol{j}$
$\boldsymbol{a_{ij}}$	Feasibility parameter; equals 1 if task $\boldsymbol{i}$ can be processed by equipment $\boldsymbol{j}$, and 0 otherwise
$\boldsymbol{CT}$	Cycle-time limit of each workstation
$\boldsymbol{EC_j}$	Investment cost of equipment type $\boldsymbol{j}$
$\boldsymbol{\tau_j}$	Reconfiguration-time parameter of equipment type $\boldsymbol{j}$; fixed in the classical MILP and ML-predicted in the proposed MILP
$\boldsymbol{CR}$	Unit cost rate associated with reconfiguration time
$\boldsymbol{\gamma}$	Task-disruption penalty coefficient
$\boldsymbol{x_{ik}^0}$	Initial assignment of task $\boldsymbol{i}$ to workstation $\boldsymbol{k}$
$\boldsymbol{y_{jk}^0}$	Initial assignment of equipment type j to workstation $\boldsymbol{k}$
$\boldsymbol{s_k}$	Ordered numerical index of workstation $\boldsymbol{k}$
Variables	
$\boldsymbol{x_{ijk}}$	Binary variable equal to 1 if task $\boldsymbol{i}$ is assigned to equipment $\boldsymbol{j}$ at workstation $\boldsymbol{k}$
$\boldsymbol{y_{jk}}$	Binary variable equal to 1 if equipment type $\boldsymbol{j}$ is installed at workstation $\boldsymbol{k}$
$\boldsymbol{z_{jk}}$	Binary variable equal to 1 if equipment type $\boldsymbol{j}$ is newly activated at workstation $\boldsymbol{k}$ for time-related reconfiguration cost
$\boldsymbol{w_i}$	Binary variable equal to 1 if task $\boldsymbol{i}$ is moved from its initial workstation
$\boldsymbol{d_{jk}}$	Binary variable equal to 1 if equipment type $\boldsymbol{j}$ is newly installed at workstation $\boldsymbol{k}$

$$\min Z = \sum_{j\in J}\sum_{k\in K} EC_j d_{jk} + CR \sum_{j\in J}\sum_{k\in K} \tau_j z_{jk} + \gamma \sum_{i\in I} w_i \quad (3)$$

$s.t.$

$$\sum_{j\in J}\sum_{k\in K} x_{ijk} = 1 \qquad \forall i \in I \quad (4)$$

$$x_{ijk} \le a_{ij} \qquad \forall i \in I, j \in J, k \in K \quad (5)$$

$$x_{ijk} \le y_{jk} \qquad \forall i \in I, j \in J, k \in K \quad (6)$$

$$\sum_{i\in I}\sum_{j\in J} t_{ij}\, x_{ijk} \le CT \qquad \forall k \in K \quad (7)$$

$$\sum_{j\in J}\sum_{k\in K} s_k x_{ijk} \le \sum_{j\in J}\sum_{k\in K} s_k x_{hjk} \qquad \forall (i,h) \in P \quad (8)$$

$$z_{jk} \ge y_{jk} - y_{jk}^0 \qquad \forall j \in J, k \in K \quad (9)$$

$$z_{jk} \le y_{jk} \qquad \forall j \in J, k \in K \quad (10)$$

$$z_{jk} \le 1 - y_{jk}^0 \qquad \forall j \in J, k \in K \quad (11)$$

$$w_i \ge \sum_{j\in J} x_{ijk} - x_{ik}^0 \qquad \forall i \in I, k \in K \quad (12)$$

$$d_{jk} \ge y_{jk} - y_{jk}^0 \qquad \forall j \in J, k \in K \quad (13)$$

$$d_{jk} \le y_{jk} \qquad \forall j \in J, k \in K \quad (14)$$

$$d_{jk} \le 1 - y_{jk}^0 \qquad \forall j \in J, k \in K \quad (15)$$

$$x_{ijk}, y_{jk}, z_{jk}, d_{jk}, w_i \in \{0,1\} \qquad \forall i \in I, j \in J, k \in K \quad (16)$$